Signals and Systems
SET 214
Lecture (09)- Fourier Transform

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Outlines

- Fourier Transform
- Exercise
- Fourier Transform Summary
- Fourier Transform Properties

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Fourier Transform

- Real world signals are rarely periodic.
- A periodic continuous-time signal can be represented in frequency domain using Fourier series.
- Transformation is the process in which either a time domain signal is converted to frequency domain or frequency domain signal is Converted to time domain so that the signal analysis becomes easy.
- Fourier Transform is an extension of Fourier Series to non-periodic signals.

$$x(t) \Longleftrightarrow X(w)$$

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$

Time domain to frequency

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w)\, e^{jwt} dw$$

Frequency domain to time

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Exercise (1/3)

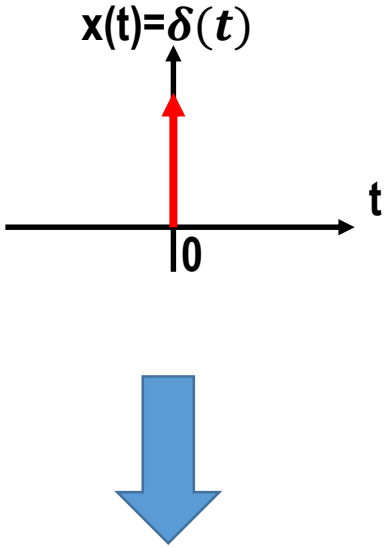
Q1) Determine the Fourier transform of: x(t)=$\delta(t)$

Ans.

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$

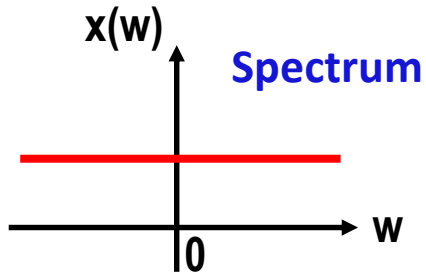
$$X(w) = \int_{-\infty}^{\infty} \delta(t)\ e^{-jwt} dt$$

$$X(w) = e^{-jw*0}=1$$



$$x(t) \Longleftrightarrow X(w)$$

$$\delta(t) \qquad 1$$



Spectrum *X (w)*

Exercise (2/3)

Q2) Determine the Fourier transform of: x(t)= $e^{-at}u(t), where\ a > 0$

Ans.

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$

$$X(w) = \int_{-\infty}^{\infty} e^{-at}u(t)\, e^{-jwt} dt$$

$$X(w) = \int_{0}^{\infty} e^{-at}\, e^{-jwt} dt$$

$$X(w) = \int_{0}^{\infty} e^{-(a+jw)t} dt$$

$$X(w) = \frac{e^{-(a+jw)t}}{-(a+jw)}\Big|_0^{\infty}$$

$$X(w) = \frac{e^{-(a+jw)*\infty} - e^{-(a+jw)*0}}{-(a+jw)}$$

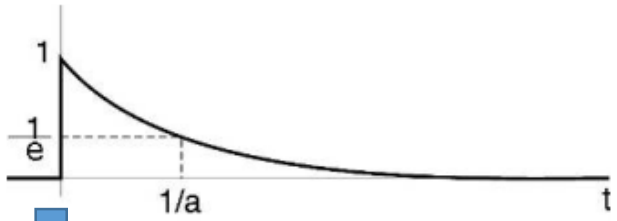
$$X(w) = \frac{0-1}{-(a+jw)}$$

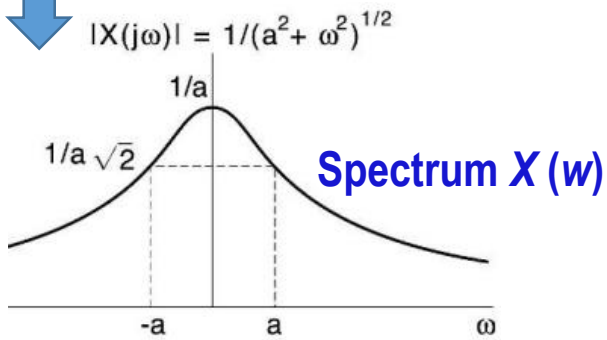
$$X(w) = \frac{1}{(a+jw)}$$

$$x(t) \Longleftrightarrow X(w)$$

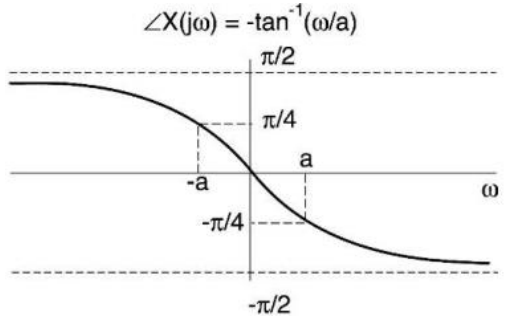
$$e^{-at}u(t) \qquad \frac{1}{(a+jw)}$$

x(t)= $e^{-at}u(t)$





Spectrum *X (w)*



Exercise (3/3)

Q3) Find the Fourier transform of the shown rectangular pulse signal

$$x(t) = \begin{cases} A, & \frac{-T}{2} \le t \le \frac{T}{2} \\ 0, & otherwise \end{cases}$$

Ans.

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$

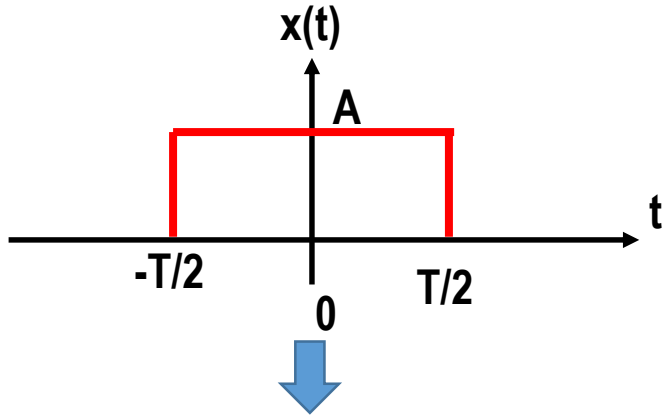
$$X(w) = \int_{-T/2}^{T/2} A\, e^{-jwt} dt$$

$$X(w) = A \left. \frac{e^{-jwt}}{-jw} \right|_{-T/2}^{T/2}$$

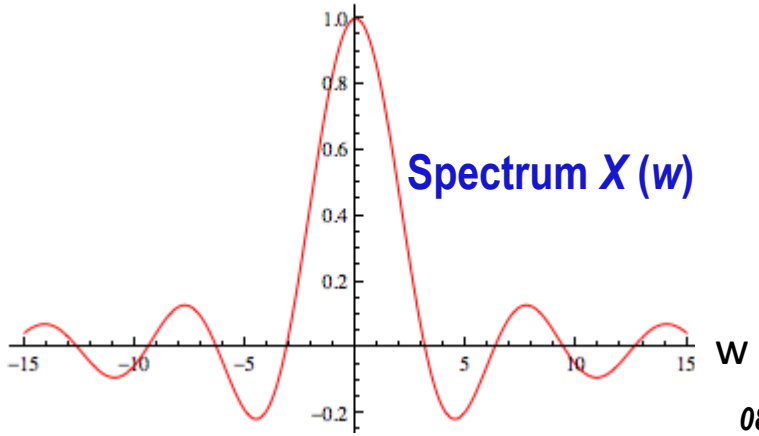
$$X(w) = A * \frac{e^{-jwT/2} - e^{jwT/2}}{-jw}$$

$$x(t) \Longleftrightarrow X(w)$$

$$A \qquad AT * Sinc(\mathrm{WT}/2)$$



sinc(w)



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Fourier Transform Summary

Time Domain	Frequency Domain
$\delta(t)$	1
$A * \delta(t)$	A
$e^{-at}u(t)$	$\frac{1}{(a+jw)}$
$x(t) = \begin{cases} 1, & \frac{-T}{2} \le t \le \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$	$AT * \text{sinc}(\frac{wT}{2})$

$$x(t) \Longleftrightarrow X(w)$$

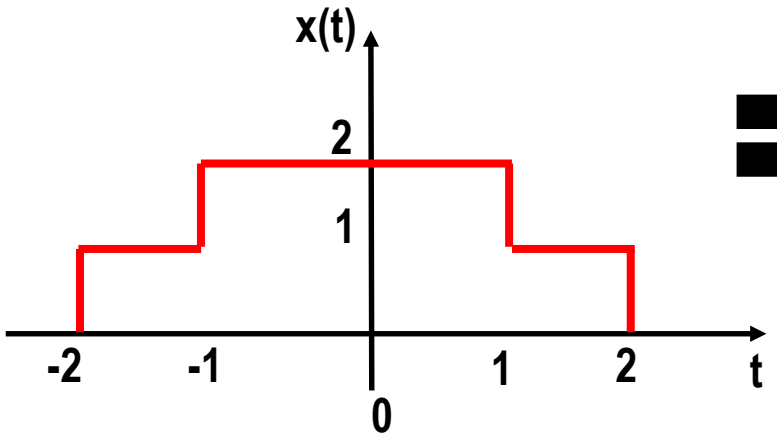
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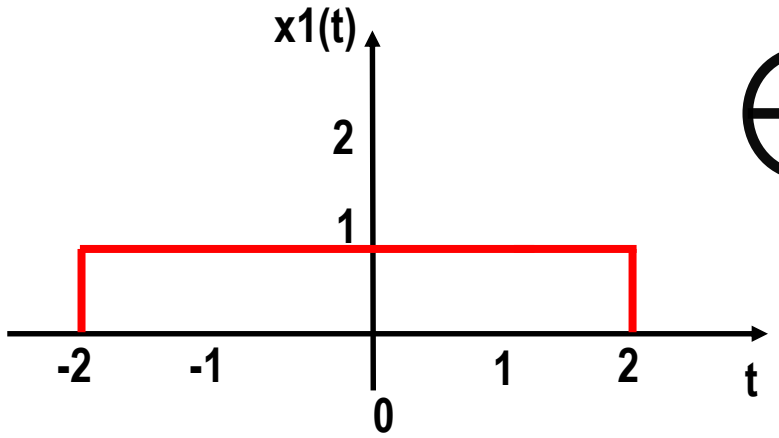
Fourier Transform Properties

Linearity ➡ $x1(t) + x2(t) \leftrightarrow X1(w) + X2(w)$

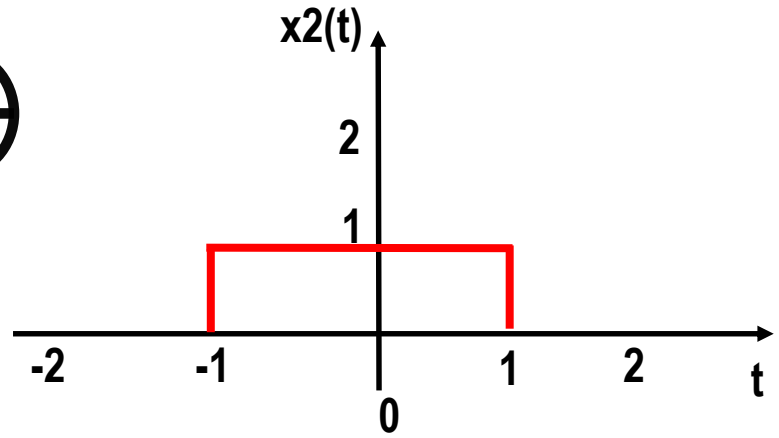
Find the Fourier Transform for x(t) using linearity property



=



⊕



$X(w) = ?$

$$X1(w) = AT * sinc(\frac{wT}{2})$$

$$X1(w) = 4 * sinc(2w)$$

$$X1(w) = AT * sinc(\frac{wT}{2})$$

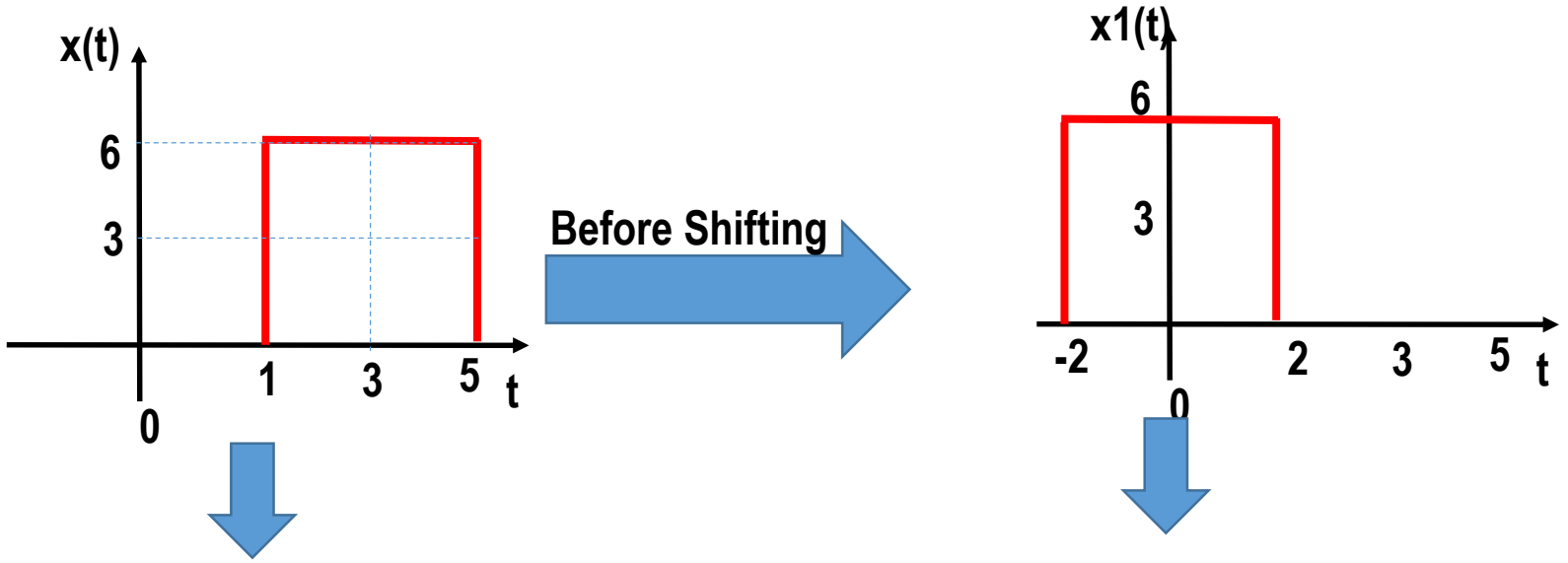
$$X1(w) = 2 * sinc(w)$$

$$X(w) = 4 * sinc(2w) + 2 * sinc(w)$$

Fourier Transform Properties

Shifting ➡ $x(t-to) \leftrightarrow X(w)\, e^{-jwt_0}$

Find the Fourier Transform for x(t) using shifting property



$X(w) = ?$

$$X1(w) = AT * sinc(\frac{wT}{2})$$

$$X1(w) = 6 * 4 * sinc(2w)$$

$$X(w) = 24 * sinc(2w)\, e^{-3jw}$$

Fourier Transform Properties

Time Scaling ➡ $x(at) \leftrightarrow \frac{1}{a} X(\frac{w}{a})$

Find the Fourier Transform for x(4t) if x(w)=$sinc(2w)$ using time scaling property

$$x(4t) \leftrightarrow \frac{1}{4}\mathrm{sinc}(2\frac{w}{4})$$

Find the Fourier Transform for x(4t-4) if x(w)=$sinc(2w)$ using time scaling and shifting properties

$$x(4t) \leftrightarrow \frac{1}{4}\mathrm{sinc}(2\frac{w}{4})$$

$$x(4(t-1)) \leftrightarrow \frac{1}{4}\mathrm{sinc}(2\frac{w}{4})\, e^{-jw}$$

Thanks!